Tommy Flowers Institute

Strategic University Research Programme Communications Special

Delivering world-class ICT research in the future will require a new breed of researcher.

We need researchers who can collaborate confidently across multiple academic disciplines - and work seamlessly between academia and industry.

That's why we in BT, along with our partners from across the ICT sector, have launched the Tommy Flowers Institute. The Institute brings the ICT industry and UK academia together to produce the research leaders of the future.

These researchers will be equipped to take on the complex challenges facing the ICT sector and help our nation to enrich its world-leading knowledge economy.

The Tommy Flowers Institute supports the RCUK (Research Councils UK) 'Research Innovate Grow' ambition to make sure the UK remains the best place on the planet to do research, innovate and grow business.

It also supports the Engineering and Physical Sciences Research Council (EPSRC's) 'accelerating impact' strategy – which is designed to maximise both academic excellence and impact through building wider and stronger research and training partnerships.



And so any organisation that seeks to make the most from collaborative technological research or has an enthusiasm to generate new research challenges need look no further than the Tommy Flowers Institute.

Fraser Burton
BT's Head of University Partnerships

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The Launch took place on 29 September 2016, at Adastral Park. We were joined by over 80 attendees from 17 Universities; 1 Vice Chancellor; 2 Pro-Vice Chancellors; 18 Professors; 7 companies; EPSRC; National physical Laboratory; Institution of Engineering and Technology; New Anglia LEP, and Suffolk County Council.



Tim Whitley, MD Research & Innovation and MD of Adastral Park opened the launch, saying "It's appropriate that the institute is named after Tommy Flowers. He brought together the best of industry and academia to create the world's first electronic programmable computer.

"This institute will bridge the gap between industrial research and the fantastic talent that exists in the academic sector."

Location is critical. Adastral Park is a major ICT innovation hub that is home to the Innovation Martlesham cluster of 85 ICT companies ranging from start-ups to global multinationals. The Park is home to BT's R&D headquarters and BT's network control centres for half the UK broadband network and the whole of its international network across 178 countries. In addition the Innovation Martlesham cluster contains many of the suppliers to the wider UK and global telecoms sector including Cisco, Intel, Huawei and Nokia. It also hosts many SMEs with world-leading technology that have the potential to grow significantly given the right facilities and support, e.g. Silicon Safe. There are over 3000 BT employees on site and over 700 employees of partner companies.

Adastral Park was recently highlighted in a special Technology article with the BBC looking at Tech Talent and the UK's digital clusters. Ipswich stood out thanks to the fact that many firms have congregated around the BT Research Headquarters at Adastral Park.

Our vision for the Tommy Flowers Institute is to have ICT industry partners engaging with UK academia on specific industrial and societal challenges, to produce both excellent research solutions and the excellent research leaders that are able to drive implementation

By 2020, the Tommy Flowers Institute will be the major national centre for supporting the development of future leaders in ICT research.

The Institute will be a growth engine for the nation's ICT sector - producing high calibre individuals to be leaders of ICT research and innovation in the UK

Bigger than any one company or any one university, the Institute will be a source of well-rounded ICT research leaders with a thorough commercial and operational perspective to complement their technical excellence.



- Bigger than any one company or any one university
- Addressing the big challenges facing the UK ICT industry

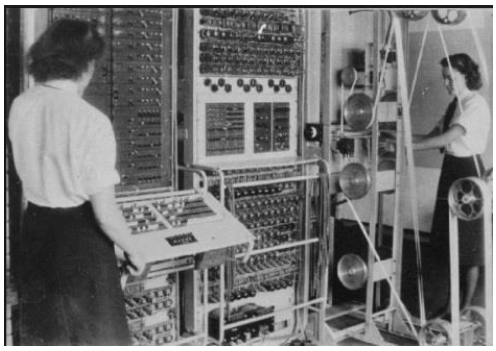




- A source of well-rounded ICT research leaders
- Deep commercial and operational understanding



Tommy Flowers' legacy illustrates how the combination of great minds from diverse backgrounds has far-reaching potential.



The opportunity for his achievement with Colossus was made possible thanks to the work of two other men - Bill Tutte and Max Newman.

Alan Turing, who broke the famous Enigma code, recognised Tommy Flowers' talent and introduced him to Max Newman, a Cambridge mathematician, who at that time was creating the Heath Robinson Machine to mechanise Bill Tutte's process of breaking the Lorenz code.

Flowers' saw the flaws in this machine and his unique outlook led him to create Colossus, the world's first electronic, programmable computer which is credited with shortening the Second World War by two years.

A Programme of "real-world" conferences.



The Institute exists to improve university researcher's employability, in both industry and academia, and thereby enable them to achieve more economic and societal impact, both as individuals and from their research.

The Institute is complementary to what academia can provide, bringing in industrial exposure, opportunities and contacts that just are not available elsewhere, at least not all in one place.

At the heart of our first year, will be exposure to strategic industry challenges with a programme of "real-world" conferences from industrial practitioners at the leading-edge of operations, technology and innovation. There will be three of these, each lasting three days, focussing not on technology, but on the challenges that the technology is being used to address, and focussing not on the research, but on what is needed from the researchers to meet those challenges.

The headline conferences will be bridged by ongoing activities throughout the year, to build on the development needs and opportunities that are identified as the output from the conference series.

The Future Organisation: how to build agility and customer focus?

This topic was chosen first because it is relevant to all stakeholders, and clearly lies beyond any one academic discipline. Technologies such as IoT and Big Data may arise from the discussions, but we are not focussing on technical research topics, but the bigger industrial challenges.

The second conference is to be on the Converged User Experience: how to deliver the connectivity and performance required? This is now becoming more telecoms-oriented and is clearly central to any business in the ICT sector. It's not about specific network technologies, but the user experience we are aiming to deliver

The third conference is the perennial challenge of Security and Trust: how to both empower and protect all users. Our users range from individual consumers to government agencies

The Conferences will run in February, June and September 2017. At the conclusion of Day 3, we hope the post-graduate students and the post-doctoral researchers will have got some valuable exposure, made use of the networking opportunities and have a better idea of the impact both they and their research can have, beyond their current academic studies.



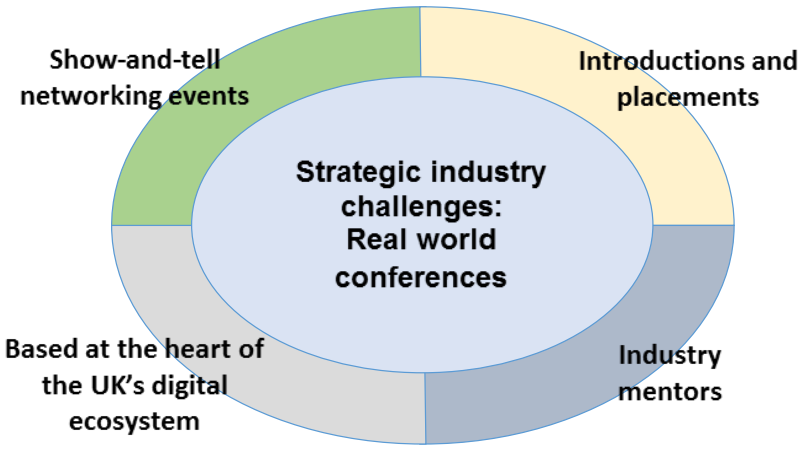
Customised Support. To keep the momentum going between conferences, but only if the researchers want it, we will provide customised support.

With "Show-and-tell" networking events, we will provide student-initiated networking events – a chance to explain your work to other researchers and companies, identify how it addresses industrial challenges

We can broker introductions between industry projects and prospective interns, leaving the bilateral terms to the two parties

We can seek out industry mentors, as requested by researchers, to meet development needs beyond their technical research

And we can provide the opportunity to be based at Adastral to work with other students and a thriving industrial community







The Researchers' View.

The Tommy Flowers Institute is a cross-industry initiative to develop ICT doctoral and post-doctoral researchers' skills to be future research leaders.

But what do today's researchers really want? Three early-career researchers discussed their thoughts at the launch.

Sam Tickle – Lancaster University PGR

Reginald Ankrah - Robert Gordon University PGR

Anasol Pena Rios - University of Essex PDRA



Having worked in industry with other ICT companies before starting his PhD, Sam felt that receiving practical and technical skills via the Tommy Flowers Institute has enormous potential. "Industry really can be a pioneer", he said.

Following 3 years in industry after graduating, Reginald received full funding from Ghana to enable him to study for his PhD. Reginald is looking for personal development skills and feels that networking with industry will really help him develop his knowledge in areas such as Innovation; skills to become a research leader and project management skills.



Anasol studied for her PhD to challenge herself, following working in the Banking and Retail industry. "You want your research to matter", said Anasol, "you want to apply your knowledge into the real world". She felt that the Tommy Flowers Institute will help researchers to do this. Help and support from Tommy Flowers Institute will enable research to move forward and make a difference.

What came through from all of the students was despite receiving good support from the University mentors, studying for a PhD can be a lonely process. They all felt that the Tommy Flowers Institute will provide help, support, and mentoring which will be very valuable.



The Industry's View. During the launch, a panel from Industry provided their view of the Institute and challenges ahead.

Richard Burguete from the National Physical Laboratory

Wenbing Yao from Huawei

Jorg-Peter Elbers from Adva

Dennis Moynihan from EIT Digital London



Richard is Director of the NPL Post Graduate Institute and fully supported the Tommy Flowers Institute. Working together will help enhance the experience of getting the students to really understand what's out there. Establishing the link between Tommy Flowers Institute and NPL's Post Graduate Institute, he believes, will be a good transfer of knowledge.



Director of Strategy and Marketing at Huawei, Wenbing Yao wished she had had more visibility about what industry needed whilst studying. She fully supported the Tommy Flowers Institute, offering Huawei as a partner. Wenbing felt that industry should let academia know what their real problems are, so that they could work together to solve problems. An area she highlighted was Virtual Reality which covered anything from CLOUD, to the pipe and keeping the information flowing through.



Jorg-Peter from Adva - an optical networking company and research partner with BT – felt that the Tommy Flowers Institute framework should be a great inspiration for researchers. Jorg-Peter felt that the networking opportunities would be great. He felt the research challenges would include processing generator updates – both the technical challenges and the economic challenges. "The ICT landscape has changed so much", he said, "from Internet of Things to Smart Cities, the company of the future has to have computing capabilities". Jorg-Peter felt that to be a future researcher, you need to be curious and passionate.



Dennis Moynihan from EIT Digital London - a leading European open innovation organisation - explained that he has partnerships across Europe in his organisation, with links into the doctoral schools, which links into the Tommy Flowers Institute really well.

"Tommy Flowers Institute is a great reminder to step back and focus on science", he said, "BT takes itself seriously for Innovation". Dennis felt that by collaborating, this drives more momentum together. He confirmed that EIT Digital London will contribute to the seminar series. "It's time to get mud on your boots", he said, "time to get dirty and get experience with real problems".

The Tommy Flowers Institute is a new way of collaborating between industry and academia: it's researcher-focussed, not just research focussed.

Both industry and academia need their future research leaders to have the greatest impact that they possibly can, and if we can work together to do this with early-career researchers now, then this can only benefit both us as a community and the UK economy overall.

This is something new: it's being kick-started by BT initially to create better research leaders for the whole ICT industry, for academia and the UK economy.

For Further information, please email tommy.flowers.institute@bt.com



The Tommy Flowers Institute Launch, 29 September 2016